

Industrial Internship Report on "Music Player with Playlist Management System"

Prepared by

Srushti S M

Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. I had to finish the project including the report in 6 weeks' time.

My project was Music Player with Playlist Management System using Core Java, Swing, JDBC, and SQLite database. The main objective of the project is to provide users with a simple and interactive music player application that allows users to import songs, create playlists, add or remove songs from playlists, and control music playback using play, pause, and stop functionalities.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

During the six weeks of the Industrial Internship Program, I worked on the project “Music Player with Playlist Management System using Core Java, Swing, JDBC, and SQLite”. The internship helped me improve my programming, problem-solving, and application development skills.

The project was developed to allow users to import songs, create playlists, add or remove songs from playlists, and control music playback using play, pause, and stop functionalities.

The opportunity provided by upskill Campus (USC) and UniConverge Technologies Pvt Ltd (UCT) gave me practical exposure to real-time software development and project implementation.

The program was planned in stages



Through this internship, I learned Core Java concepts, Swing GUI development, JDBC database connectivity, exception handling, and debugging techniques. Overall, it was a valuable learning experience for my career growth.

I would like to thank upskill Campus, UniConverge Technologies Pvt Ltd (UCT), The IoT Academy, family members, and friends for their support and guidance throughout the internship.

My message to juniors and peers is to participate actively in internships and practical projects because they help improve technical knowledge, confidence, and industry readiness.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



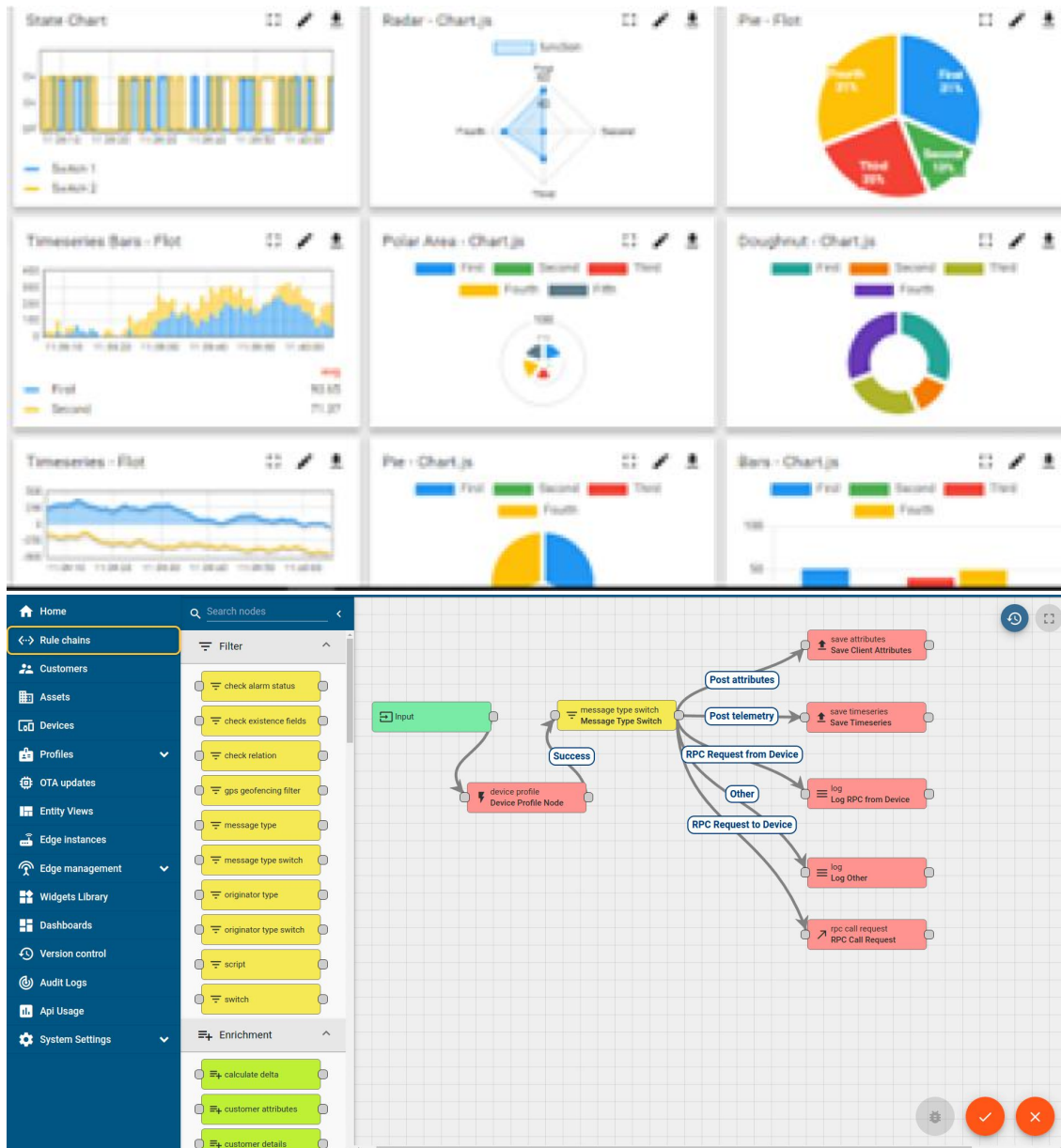
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



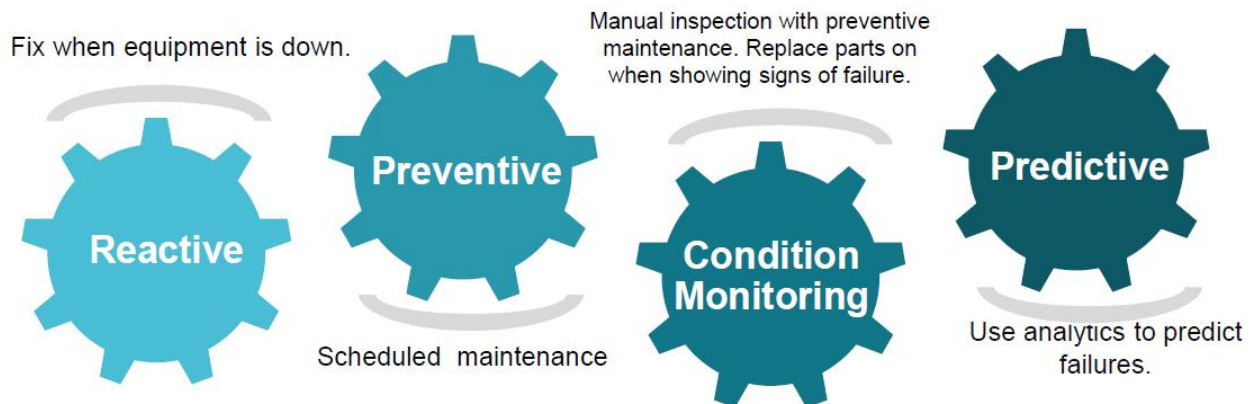


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

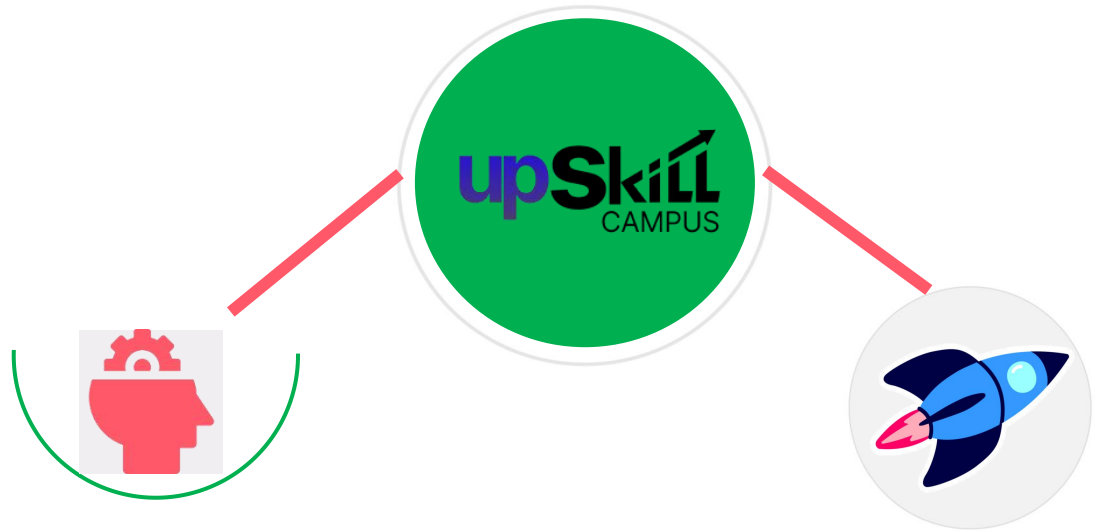
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

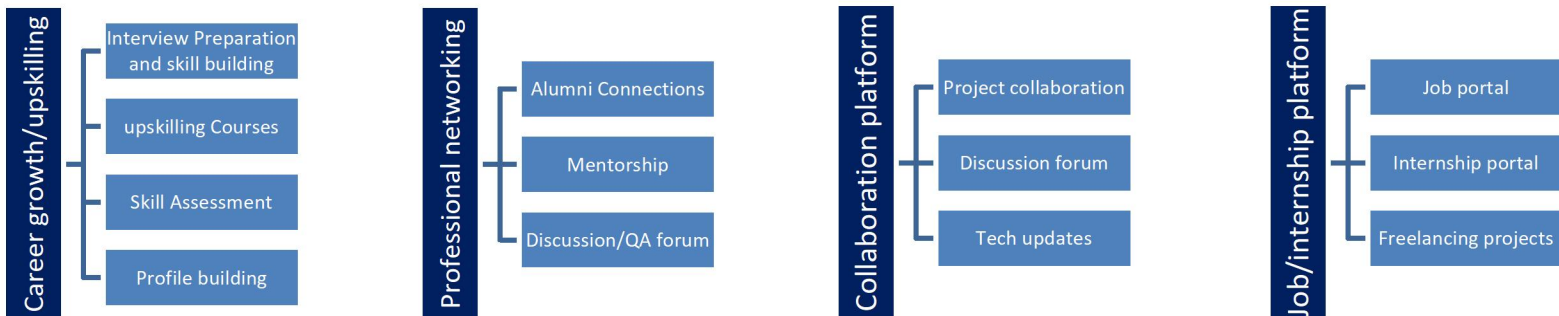
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The main objectives are:

- To develop a user-friendly graphical music player using Java Swing.
- To implement music playback controls such as Play, Pause, and Stop.
- To store and manage song information using JDBC and SQLite database.
- To create and manage playlists dynamically.
- To improve practical knowledge in Core Java and database connectivity.

2.5 Reference

[1] Oracle Java Documentation

[2] SQLite JDBC Documentation

[3] Eclipse IDE Documentation

2.6 Glossary

Terms	Acronym
JDBC	Java Database Connectivity
GUI	Graphical User Interface
SQLite	Lightweight Relational Database
Swing	Java GUI Framework
WAV	Audio File Format

3 Problem Statement

In the assigned problem statement

The assigned problem statement is to develop a Music Player with Playlist Management System using Core Java, Java Swing, JDBC, and SQLite database.

The application should allow users to import and manage music files efficiently. Users should be able to create playlists, add or remove songs from playlists, and perform music playback operations such as Play, Pause, and Stop.

The system should store song details and playlist information in a database using JDBC connectivity. The project also focuses on providing a user-friendly graphical interface for smooth interaction and efficient playlist management.

The main purpose of this project is to understand practical implementation of Core Java concepts, GUI development, database connectivity, file handling, and event-driven programming in real-time applications.

4 Existing and Proposed solution

Existing music player applications often require internet access, consume more system resources, and include unnecessary complex features.

The proposed solution is a lightweight desktop-based Music Player application developed using Core Java technologies. The application provides simple playlist management and music playback features using Java Swing and JDBC.

Advantages of the proposed system:

- Simple and easy-to-use interface
- Efficient playlist management
- Lightweight application
- Database support using SQLite
- Easy song import and playback

4.1 Code submission (Github link) : <https://github.com/Srushtism/UpskillCampus.git>

4.2 Report submission (Github link) : <https://github.com/Srushtism/UpskillCampus.git>

5 Proposed Design/ Model

The Music Player system is designed using Java Swing for GUI components and SQLite database for storing song and playlist details.

The system consists of the following modules:

- Song Import Module
- Playlist Management Module
- Music Playback Module
- Database Connectivity Module
- GUI Interaction Module

The user interacts with the graphical interface to manage songs and playlists. JDBC is used to communicate with the SQLite database for storing and retrieving data.

5.1 High Level Diagram

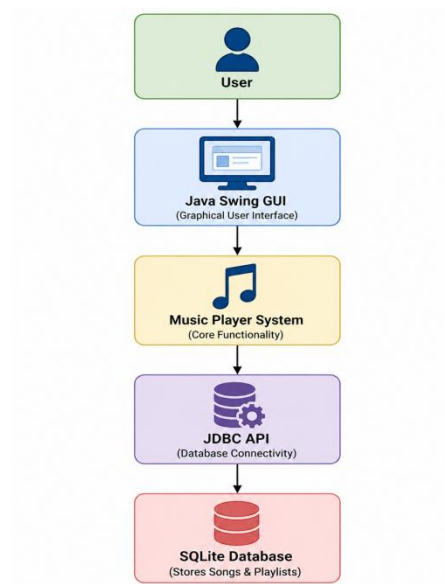


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

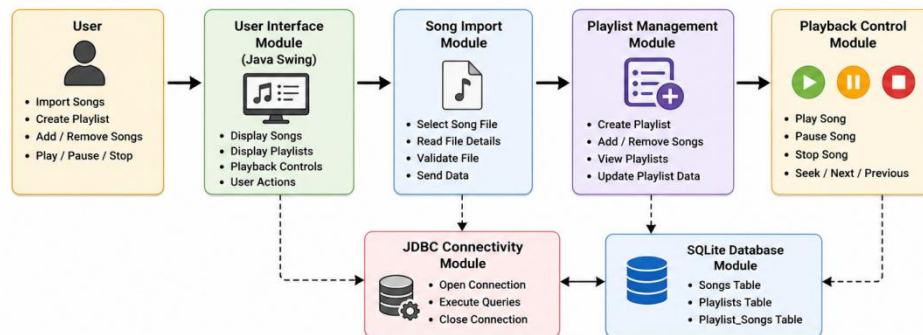
5.2 Low Level Diagram



Figure 2: LOW LEVEL DIAGRAM OF THE SYSTEM

5.3 Interfaces

Block Diagram



Flow Chart

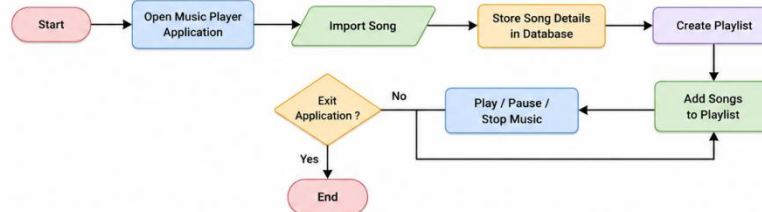
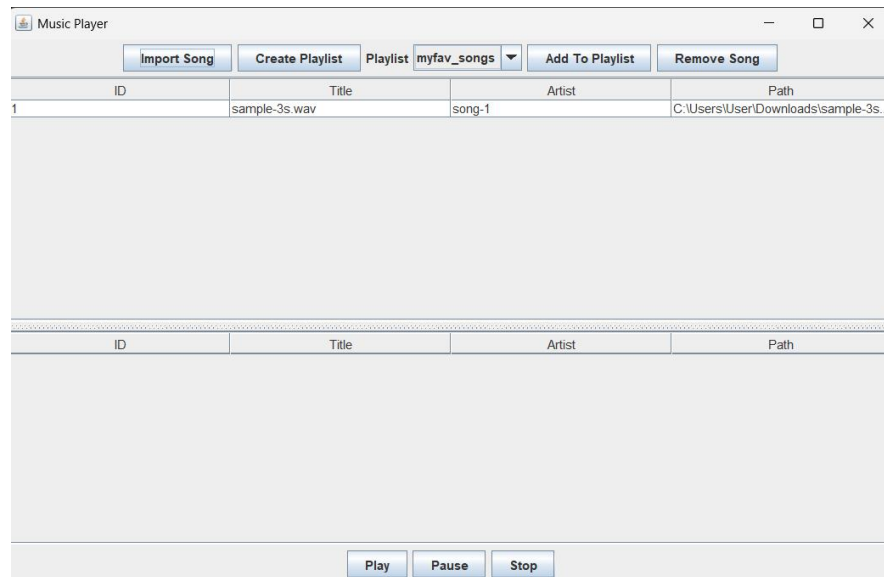
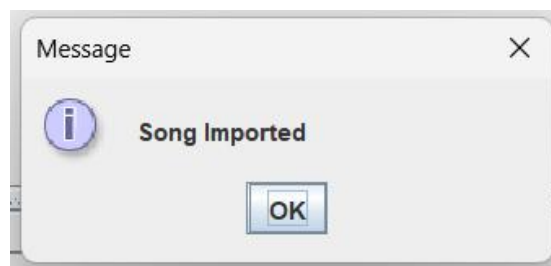


Figure 3: BLOCK DIAGRAM AND FLOW CHART OF THE SYSTEM

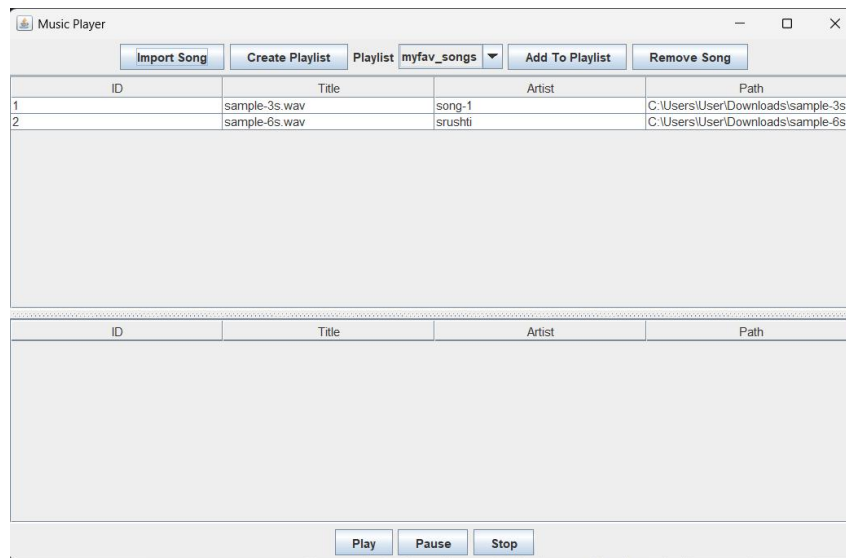
5.4 Screenshots



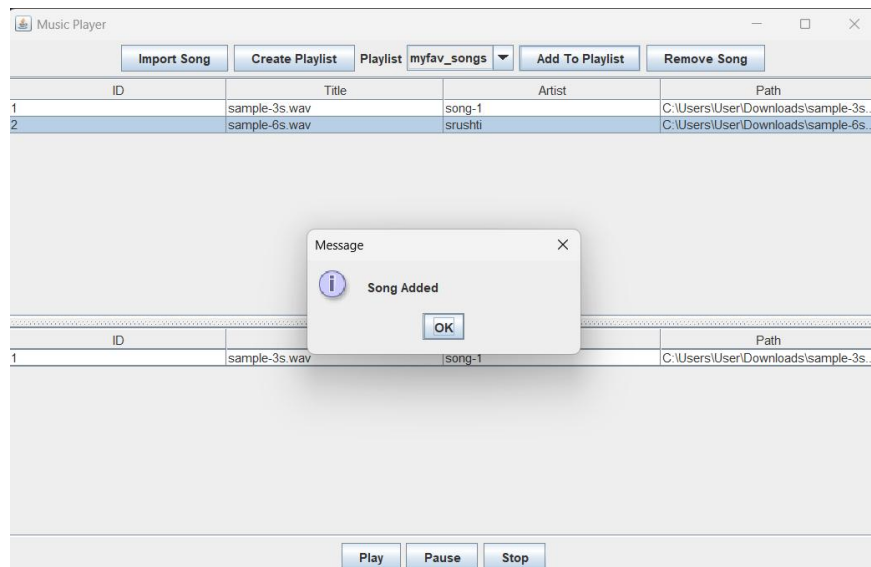
5.4.1: import song page



5.4.2: Song Imported page



5.4.3: Playlist created



5.4.4: Song added to Playlist

6 Performance Test

The application was tested with different song files and playlist operations.

The system successfully handled:

- Song importing
- Playlist creation
- Song addition/removal
- Music playback controls

The application performed efficiently with minimal memory usage and provided smooth user interaction.

6.1 Test Plan/ Test Cases

Test Case	Expected Result	Status
Import Song	Song added successfully	Pass
Create Playlist	Playlist created	Pass
Add Song	Song added to playlist	Pass
Play Song	Music plays	Pass
Remove Song	Song removed	Pass

6.2 Test Procedure

The Music Player with Playlist Management System was tested in Eclipse IDE using Java, Swing, JDBC, and SQLite database. The testing process was carried out to verify that all modules function correctly and meet the project requirements.

Initially, the application was launched and database connectivity was verified. The song import functionality was tested by selecting audio files and storing their details in the SQLite database. The playlist creation feature was then tested by creating multiple playlists with different names and verifying that the records were stored successfully.

Further testing was performed on playlist management operations such as adding songs to playlists, viewing playlist contents, and removing songs from playlists. Playback controls including Play, Pause, and Stop were tested multiple times to ensure smooth audio execution and proper response to user actions.

Database operations were tested by inserting, retrieving, and deleting records. Error handling mechanisms were also verified by attempting invalid operations and checking whether appropriate messages were displayed to the user. The graphical user interface was observed for responsiveness and ease of navigation during continuous usage.

The application was executed repeatedly to verify stability and ensure that no unexpected errors occurred during normal operation.

6.3 Performance Outcome

The testing results showed that the Music Player application performed successfully under normal operating conditions. Songs were imported and stored in the database correctly, and playlist management functions worked as expected.

The graphical user interface responded quickly to user actions, while SQLite provided efficient storage and retrieval of song and playlist data. Playback operations such as Play, Pause, and Stop were executed without noticeable delays.

Although advanced performance measurements were not conducted, the application demonstrated reliable functionality and stable performance for a desktop-based music management system. The results indicate that the proposed design successfully meets the project objectives and can be enhanced with additional features in the future.

7 My learnings

The internship provided me with an excellent opportunity to gain practical exposure to software development using Core Java technologies. During this internship, I worked on the project “Music Player with Playlist Management System” and learned how to develop a complete desktop application from requirement analysis to implementation and testing.

One of the major learnings from this project was improving my Core Java programming skills. I gained hands-on experience with object-oriented programming concepts such as classes, objects, inheritance, encapsulation, and exception handling. These concepts helped me build a structured and maintainable application.

I also learned Java Swing for designing graphical user interfaces. Through Swing components such as JFrame, JButton, JTable, JComboBox, JPanel, and JScrollPane, I understood how to create interactive desktop applications. Designing the user interface helped me improve my understanding of event-driven programming and user interaction.

Another important learning was database connectivity using JDBC. I learned how Java applications communicate with databases, execute SQL queries, and manage data efficiently. Using SQLite database, I gained experience in creating tables, inserting records, retrieving data, and performing update and delete operations.

The project also improved my knowledge of file handling and audio processing. I learned how to import audio files, manage file paths, and implement music playback functionalities such as Play, Pause, and Stop. Understanding how different modules interact with each other gave me a better understanding of software architecture and application design.

Throughout the development process, I faced several challenges such as database connectivity issues, debugging errors, handling exceptions, and managing application flow. Solving these problems improved my analytical thinking, troubleshooting skills, and confidence in coding.

In addition to technical skills, the internship enhanced my project management and documentation skills. I learned the importance of planning, testing, maintaining project reports, and presenting technical work in a professional manner.

Overall, this internship was a valuable learning experience that strengthened my technical knowledge in Core Java, Swing, JDBC, and database management. It also improved my problem-solving abilities, software development skills, and confidence to work on real-world industry projects in the future.

8 Future work scope

Future improvements for this project include:

- MP3 file support
- Advanced music controls
- User login system
- Online music streaming
- Dark mode interface
- Search and filtering functionality
- Volume control and equalizer support